

European Journal of Housing Policy

ISSN: 1461-6718 (Print) 1473-3269 (Online) Journal homepage: www.tandfonline.com/journals/reuj19

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To cite this article: George Galster (2007) Neighbourhood Social Mix as a Goal of Housing Policy: A Theoretical Analysis, European Journal of Housing Policy, 7:1, 19-43, DOI: 10.1080/14616710601132526

To link to this article: https://doi.org/10.1080/14616710601132526

Published online: 15 Feb 2007.

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European Journal of Housing Policy
Vol. 7, No. 1, 19–43, March 2007

Routledge
Taylor & Francis Group

Neighbourhood Social Mix as a Goal of Housing Policy: A Theoretical Analysis

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ABSTRACT *Many western European housing policies have tried to increase the residential mix of advantaged and disadvantaged groups. Unfortunately, policymakers have given little consideration to how these groups will interact as neighbours. There are numerous theoretically grounded mechanisms by which the social mix of a neighbourhood may influence socio-economic outcomes of its residents. These mechanisms differ on the basis of which group is generating the social externality in the neighbourhood, whether this externality is positive or negative, whether it affects all residents equally, and whether the marginal externality generated by adding one more member of a particular group is constant, proportional, or is characterized by a threshold effect. This paper demonstrates that a social mix housing policy can be justified only under a circumscribed set of the preceding parameters. Indeed, depending on the mechanism assumed, social efficiency implies that neighbourhoods should be either: equally mixed, have the disadvantaged group dispersed as widely as possible, or rigidly segregated; for other mechanisms, mix becomes irrelevant. Thus, for formulating and justifying a mixed housing policy on either efficiency or equity grounds it is crucial to understand exactly what sort of neighbourhood effect(s) is operating in neighbourhoods.*

KEY WORDS: Neighbourhood effects, social mix, social exclusion, European housing policy, externalities

Introduction

This paper is motivated by one salient fact. Increasingly, in many western European nations, housing policymakers see enhancing the social diversity of residential environments as an important goal, although how 'diversity' is defined differs by national context (Kleinhans, 2004; Andersson, 2006). As illustration, in the United Kingdom a variety of taskforce reports and white papers (e.g. Rogers, 1999; DoE, 2000; ODPM, 2003), buttressed by an emerging scholarly consensus (Atkinson & Kintrea, 2000, 2002; Minton, 2002; Tunstall, 2003, 2005; Berube, 2005; Tunstall & Fenton, 2006), have decried homogeneous social housing estates and contrasted them to the benefits of social inclusion and sustainability that flow from neighbourhood income and

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ISSN 1461-6718 Print/1473-3629 Online 07/010019–25 © 2007 Taylor & Francis
DOI: 10.1080/14616710601132526

tenure mixing. The dominant principle of Dutch policy regarding the integration of ethnic minority and 'socio-economically weak' groups, since the close of World War II, has been to provide social opportunities through mixed residential environments (Musterd, 2003; Musterd *et al*., 2003; Lindeman *et al*., 2003; Penninx, 2006). This approach has been echoed in the 1998 Swedish 'Development and Justice' policy aimed at combating economic, social and ethnic segregation (Andersson, 2006). This emphasis on social mix typically has been justified on grounds of both economic efficiency (e.g. making society as a whole better off by enhancing solidarity, labour productivity and community sustainability) and distributive equity (e.g. improving the life-chances and social inclusion of disadvantaged groups); see, e.g., Delorenzi (2006).

Several programmatic responses are indicative of this thrust towards social mixing, again with particulars varying by national context. In Sweden, France, the UK and the Netherlands, for example, there have been widespread, large-scale investments aimed at restructuring large, homogeneous, post-war neighbourhoods and housing estates (through selective demolition, infill construction and sale of social housing) so that they contain a greater diversity of housing types by price range and tenure (Atkinson & Kintrea, 2000, 2002; Dekker & van Kempen, 2004; van Kempen *et al*., 2005; Turkington *et al*., 2004). In the UK we have seen an emphasis on both 'right to buy' and 'choice-based letting' in social housing, and substantial changes in local planning policies to stimulate development of new mixed neighbourhoods (Minton, 2002; Martin & Wilkinson, 2003; Tunstall, 2005; Kearns & Mason, 2005). In the Netherlands it is now required that new, larger-scale residential developments must set aside a minimum share of the dwelling units for social housing. And in Sweden we see a long-standing policy of dispersing new immigrants being supplemented by a new Metropolitan Development Initiative, which seeks to reduce segregation by making targeted neighbourhoods more attractive to homeseekers with many options (Edin *et al*., 2003; Andersson, 2006).

Despite this widespread policy thrust, the premise that neighbourhood social mix is a desirable goal on either efficiency or equity grounds has not been rigorously supported theoretically or empirically. Indeed, the premise has been recently challenged (Ostendorf *et al*., 2001; Musterd, 2002, 2003; Musterd *et al*., 2003; Delorenzi, 2006) and has proven controversial for an extended period (Sarkissian, 1976; Cole & Goodchild, 2001).

My purpose in this paper is to contribute a critical analytical perspective on this policy-relevant issue of neighbourhood social mix. Specifically, I aim to: (1) delineate comprehensively the various potential ways in which neighbourhoods might affect residents' socio-economic outcomes; (2) present a theoretical analysis that derives optimal levels of social mix based on each of the aforementioned neighbourhood effect mechanisms; and (3) draw inferences for housing policy.

I posit that there are numerous plausible, theoretically grounded mechanisms by which the social mix of a neighbourhood may have influence on the social outcomes of its residents. These mechanisms differ on the basis of which group is generating

the inter-group social externality in the neighbourhood, whether this externality is positive or negative, whether it affects all residents equally, and whether the marginal externality generated by adding one more member of a particular group is constant, proportional or is characterized by a threshold effect.

This parsing is not mere pedantry, for each type holds very different implications for whether and how much neighbourhoods should be mixed, based on either social equity and/or efficiency rationale, as I will show in this paper. Depending on the mechanism assumed, I demonstrate that neighburhoods should be either: equally mixed; have the disadvantaged group dispersed as widely as possible; or rigidly segregated. For still other mechanisms, mix becomes irrelevant. Thus, for formulating and justifying a mixed housing policy it is crucial to understand exactly what sort(s) of neighbourhood effect(s) is operating.

This paper is organized in three major sections as follow. I begin by briefly reviewing theories of different mechanisms through which 'neighbourhood effects' may transpire. Second, I analyse in detail 11 realistic alternatives for how neighbourhood effects might occur, showing both graphically and in a mathematical appendix what optimal social mixing outcome each implies. Third, I conclude by drawing implications of the analysis for western European housing and social policymakers.

Alternative Mechanisms of Neighbourhood Effects

What occurs in neighbourhoods to potentially generate behavioural effects about which policymakers might be concerned? There have been several comprehensive reviews of the theoretical links between neighbourhood processes and individual outcomes; see especially Jencks and Mayer (1990), Duncan *et al*. (1997), Gephart (1997), Friedrichs (1998), Atkinson *et al*. (2001), Haurin *et al*. (2002), Sampson *et al*. (2002) and Ioannides and Loury (2004). I therefore will list these mechanisms and describe them only briefly here.

Neighbourhood effects arising from internal social interrelationships

One set of potential neighbourhood effects occurs when the characteristics, behaviours or attitudes of one neighbourhood resident has a direct influence on (at least a portion of) his or her neighbours. This mechanism can be thought of as a *social externality* . . . a concept that I will employ extensively below in analysing the policy implications of these mechanisms. Numerous possibilities here have been forwarded:[1]

- *Socialization:* behaviours and attitudes may be changed (for better or worse) by contact with peers or role models who may be neighbours. When these changes occur they are often referred to as 'contagion effects'. For example, the actions by some youths to vandalize common neighbourhood spaces may encourage others in the area to do the same. These socialization effects are modelled in Case 1 below.

- *Epidemic/social norms:* this is a special subset of socialization effects that are characterized by a minimum threshold being achieved before noticeable consequences arise. The need for some subset of the neighbourhood population to reach a critical mass before their social norms begin to influence residents is a case in point. Cases 2 and 3 below model negative and positive social externalities spread in this way; their combination is modelled in Case 4.
- *Selective socialization:* this process is another special type of socialization process wherein neighbours are not all equally affected by others. Employed residents are often viewed as positive role models encouraging (only) their unemployed neighbours to find work, for example. Conversely, secondary school dropouts may discourage only their same-age peers from attending school. Analogues of these processes are portrayed in Cases 5 and 6 below.
- *Social networks:* although one may say that socialization proceeds through social networks, we specify this as a distinct process involving the interpersonal communication of information and resources. One local group may intensify the density and multi-nodal structure of their social networks (create 'strong ties') by clustering, thereby increasing the sources of assistance in times of need. On the other hand, such situations may lack the 'weak ties' that offer the prospect of bringing new information and resources into the community, thereby increasing social isolation. These phenomena are explored in Cases 7 and 8.
- *Competition:* under the premise that certain local resources are limited and not public goods, this theory posits that groups within the neighbourhood will compete for these resources among themselves. Because the context is a zero-sum game, social conflict will arise as one group more successfully competes. The control of a local public park for the specialized activities of one group provides one example. Case 9 explores this mechanism.
- *Relative deprivation:* this mechanism suggests that residents who have achieved some socio-economic success will be a source of disamenities for their less well-off neighbours. The latter will view the successful with envy or will make them perceive their own relative inferiority as a source of dissatisfaction. This can be modelled analogously as competition theory, and thus is also portrayed in Case 9.
- *Stigmatization:* stigmatization of a place transpires when important institutional, governmental or market actors negatively stereotype all residents of a place and/or reduce the flows of resources flowing into the place because of its household composition. This might occur as the percentage of households in some disadvantaged ethnic group in the neighbourhood exceeds the threshold at which they are perceived by these external actors as 'dominant'. This is modelled in Case 10.

Neighbourhood effects arising from external sources

Another set of reputed neighbourhood effect mechanisms is determined by larger structural forces in the metropolitan area that are external to the neighbourhood.

Several such mechanisms have been forwarded in the literature (e.g. Bauder, 2001):

- *Spatial mismatch:* certain neighbourhoods have little accessibility (in either spatial proximity or as mediated by transportation networks) to job opportunities appropriate to the skills of their residents.
- *Local institutional resources:* certain neighbourhoods have access to few private, non-profit, or public institutions and organizations.
- *Public services:* certain neighbourhoods are located within local political jurisdictions that offer inferior services and facilities.

These externally arising mechanisms will be discussed collectively in Case 11 below.

Deconstructing Mechanisms of Neighbourhood Effects: Policy Implications for Neighbourhood Mix

In this section I systematically probe each of the aforementioned mechanisms of neighbourhood effects. I lay out explicitly their assumptions about the social externalities involved: which group is producing them, whether they are positive or negative, whether they transpire on the margin in fixed amounts, varying amounts, or after thresholds and who is being influenced by them. In each case I will derive implications on equity and efficiency grounds for what sort of neighbourhood composition is most desirable.

At the outset let me define what I mean by equity and efficiency grounds. By equity grounds I mean that, for unspecified normative reasons, a residential location pattern that disproportionately enhances the well-being of less-advantaged citizens, perhaps at the absolute sacrifice of the well-being of more-advantaged ones, is considered preferable. I adopt as a social efficiency criterion a conventional utilitarian position: try to maximize the greatest good for the greatest number of households in the given metropolitan society. In particular, below I will examine how alternative distributions of households across neighbourhoods might lead to variations in such social 'goods' as cohesion, solidarity, information, positive role models, safety, and such social 'bads' as tension, conflict, disorder and insecurity. For simplicity I assume that any such potential outcome is valued equally in utilitarian terms by both disadvantaged and advantaged households alike. Thus, if one potential allocation of households across neighbourhoods produces a situation where the (net) social goods are superior to those associated with an alternative allocation, the former is considered more socially efficient (i.e. Pareto superior).

For simplicity of exposition (but with no loss of generality), I make several assumptions. First, there are two household groups, generically labelled 'advantaged' (A) and 'disadvantaged' (D), of predetermined (not necessarily equal) numbers. It is

immaterial for this analysis on what basis households are classified into A or D groups: income, ethnicity or immigrant status. Second, these characteristics do not change during the period in question when we are assessing externality effects from mixing.[2] Third, all households within a group are identical in the extent to which they produce and/or are affected by intra-neighbourhood externalities.[3] Fourth, society consists of two neighbourhoods of equal numbers of households whose boundaries and housing stock are fixed in advance. Fifth, both neighbourhoods contain a 'quantum' of households – say a thousand – so that all expressions of parameters can be thought of as 'per thousand' without adding cumbersome labels. In this framework it is immaterial whether one thinks of alternative neighbourhood mixes as differing in the number or the percentage of A and D households present. Last, I assume no spatial spillovers of externalities between neighbourhoods; all externalities are intra-neighbourhood here.

Let the intra-neighbourhood behavioural and attitudinal externalities (i.e. changes in behaviours or attitudes of neighbours associated with various mixtures of A and D households in the neighbourhood) be summarized by an index I, that can assume positive values (good external effects) and negative values (bad external effects) compared to the baseline situation (where I is normalized to zero). The index's total value (I_T) is the sum of the externalities generated by the allocation of A across both neighbourhoods (I_A) and the externalities generated by D across both neighbourhoods (I_D). Without loss of generality I express the I_T functions in terms of percentage of D households in one neighbourhood (%D), which means in this neighbourhood there are (100-%D) A households, and in the other neighbourhood there are (100-%D) D households and (%D) A households (again, with all units in thousands).

To maximize efficiency, assume that policymakers wish to achieve the household mix in both neighbourhoods that maximizes I_T (or minimizes it when I_T is negative). To improve equity (again assuming a desire to redistribute well-being from A to D, at least to some extent), assume that policymakers wish to allocate households in ways that neighbourhood(s) with net positive externalities are not merely inhabited by A and neighbourhood(s) with net negative externalities are not only occupied by D.

The analysis that follows is comparative static in nature: I considers various *de novo* allocations of A and D households across these two hypothetical neighbourhoods, not dynamic processes of transforming pre-existing allocations. That is, I analyse the comparative social consequences of various alternative allocations of fixed amounts of A and D across a two identical neighbourhoods with fixed and equal total households in all cases. The discussion below proceeds with a reliance on graphic exposition; a parallel mathematical exposition is provided in the Appendix. With these bases established, I turn to alternative cases that exhaustively describe the various types of intra-neighbourhood social externalities that have been forwarded.

Deconstructing neighbourhood effects arising from internal social interrelationships

Case 1: Group D generates constant marginal negative externality β for all neighbours, group A generates constant marginal positive externality φ for all neighbours

This case describes the 'socialization' mechanism. Each additional D household (replacing an A household in the given neighbourhood) may provide another inappropriate role model for all neighbours (A and D alike). Or, they may try to recruit all neighbours into illegal activities. Or they may engage in publicly violent acts so that neighbours fear to leave their dwellings. By contrast, each additional A household (replacing a D household in the given neighbourhood) may provide a positive socializing influence on all neighbours, such as increasing the exposure of D immigrants to mainstream culture of the host nation.

In this case, the externality functions for some representative neighbourhood may be portrayed as in Figure 1. Figure 1 plots the percentage of households in this neighbourhood in group D (%D) against the externality index (I) associated with both groups, with I normalized to zero indicative of a baseline situation with no internal neighbourhood externalities. The relationship for D is shown as I_D: a straight line beginning at the origin (D cannot generate any externality when they are not present) and negatively sloping ($= \beta$) thereafter, signifying that each added D household reduces the collective well being in the neighbourhood by β. The relationship for A is shown as I_A: a straight line beginning at α (A generate α externalities when they are the only group present) and negatively sloping ($= \gamma$) thereafter, signifying that each replacement of an A household by a D household reduces the collective well being in the neighbourhood by γ.

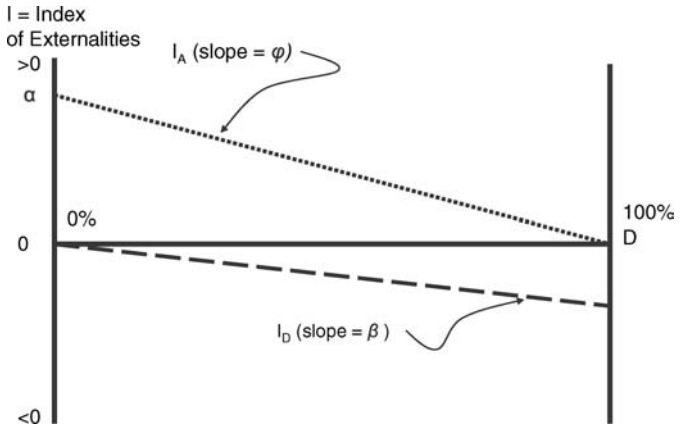


Figure 1. Case 1: A has constant marginal externality $= \varphi > 0$ for all; D has constant marginal externality $= \beta < 0$ for all.

Under these assumptions, the perhaps surprising implication is that *any mixture* of groups between neighbourhoods will produce exactly the same total amount of externalities, and thus are equally efficient using our standard. This remarkable and important conclusion is explicated more fully in Galster (2002). The intuition is as follows. Switching any group D household from one neighbourhood to another will reduce I_D by β in the origin neighbourhood and raise it by β in the destination neighbourhood, yielding no net change in aggregate. An analogous argument can be made when switching a group A household: the marginal gain of φ where one is added will be offset by the marginal loss of φ where one is subtracted. Note this conclusion holds regardless of whether A and/or D are assumed to have either positive, zero or negative externalities; so long as the group's externality is constant on the margin, efficiency will not be affected by mix.

From an equity standpoint there may be ground for mixing, however. Under the assumptions here, complete segregation would mean that all D households would be suffering from the negative externalities produced by their D neighbours, while all A households would be benefitting from the positive externalities generated by their A neighbours. A fairer sharing of the burdens and benefits across groups would suggest a fairly even mixing of groups in all neighbourhoods.

Case 2: Group D generates constant marginal negative externality β for all neighbours beyond threshold X (expressed as %D)

Here, with the 'epidemic/social norm' mechanism, the marginal neighbourhood effect is not constant but rather commences once the group generating it exceeds a critical value.[4] Altered norms via collective socialization are representative of such a process. Collective socialization theories focus on the role that social groups exert on shaping an individual's attitudes, values and behaviours (e.g. Simmel, 1971; Weber, 1978). Such an effect can occur to the degree that: (1) the individual comes in social contact with the group; and (2) the group can exert more powerful threats or inducement to conform to its positions than competing groups. These two preconditions involve the existence of a threshold. Given the importance of interpersonal contact in enforcing conformity, if the individuals constituting the group in question were scattered innocuously over urban space, they would be less likely to be able either to convey their positions effectively to others with whom they might come in contact or to exert much pressure to conform. It is only when a group reaches some critical mass of density or power over a predefined area that it is likely to become effective in shaping the behaviours of others. Past this threshold, as more members are recruited, the group's power to sanction non-conformists probably grows nonlinearly. This is especially likely when the position of the group becomes so dominant as to become normative in the area.[5]

In this instance, the group D externality function for a neighbourhood will appear as I_D in Figure 2. Until the percentage of D households exceeds the threshold X there will

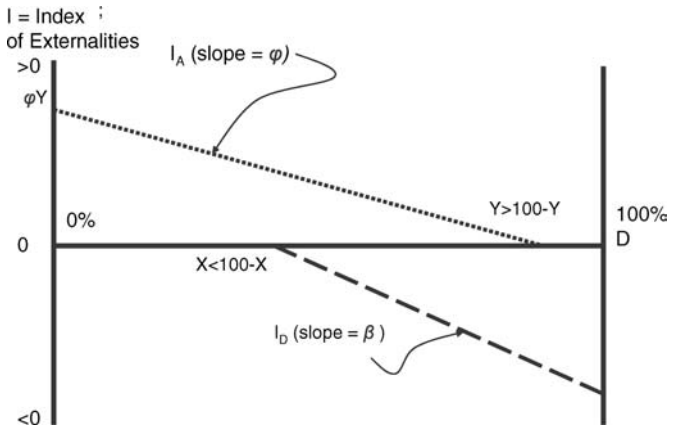


Figure 2. Case 2: D has threshold @X marginal externality $= \beta < 0$ for all. Case 3: A has threshold @Y marginal externality $= \varphi < 0$ for all. Case 4: both of the above.

be no externality manifested; thereafter each additional D imposes a constant marginal negative externality.[6] Here, it is clear that efficiency would be maximized if in every neighbourhood %D could be kept at or below X per cent of the households, for then there would be no negative externalities anywhere. This may not be possible, however, given the overall size of D households relative to the value of X. If the percentage of the entire population of households represented by D were larger than X, the decline in I_T overall would be minimized by allocating X per cent D households in as many neighbourhoods as possible, by implication leaving 100 per cent occupancy by D in the others. Thus, unlike Case 1, when there is a threshold for the neighbourhood externality there are very precise implications for neighbourhood mix strategy on efficiency grounds, which essentially have as their target a ceiling quota of the negative externality-producing group and, failing that, their complete segregation.

On equity grounds there remains a basis for mixing, comparable to that employed in Case 1. In order to avoid imposing negative externalities on D households themselves, they should be restricted from exceeding their threshold. Note the extreme paternalism implicit here: D must be limited in a neighbourhood 'for their own good'.

Case 3: Group A generates constant marginal positive externality φ for all neighbours beyond threshold Y (where Y defined as maximum %D where φ persists)

Here, with the 'epidemic/social norm' mechanism, we have the converse of Case 2, with group A producing positive externalities for all if they exceed a minimum threshold Y (i.e. %D can be no larger than Y). The process of social norm transmission operates here in identical fashion to that described above in Case 2, except that past the threshold socially desirable collective socialization processes ensue.

The efficiency analysis follows as above; see the I_A function in Figure 2. To maximize the sum of positive externalities we should avoid having neighbourhoods where group A households represent less than their threshold (i.e. < %D-Y). So long as this is achieved the allocations of group A households among these neighbourhoods will not matter since their marginal externalities are constant, as per the logic of Case 1. So, for example, if there needed to be 30 per cent of group A in a neighburhood before their externalities started, it does not matter for efficiency grounds whether all the neighbourhoods in which group A resides have 100 per cent A or 31 per cent A, so long as none have less than 30 per cent. Conceivably, therefore, this set of assumptions could imply that either extremely segregated or mixed situations for group A are equally efficient. Of course, if this threshold is very high compared to the share of group A in the overall population, it will be impossible to fulfil this 'minimum share' requirement for all neighburhoods, in which case the optimal strategy would be to fill as many neighbourhoods as possible with 100 per cent A households, with any remaining group A households distributed, however, among the rest. Thus, not only the potential existence of a threshold but its magnitude relative to the group in question becomes critical for guiding a neighbourhood mix strategy.

From an equity perspective, Case 3 suggests that achieving an above-threshold percentage of group A households would be desirable because group D would also benefit thereby. However, the complete segregation of group A would not be as desirable form an equity standpoint, since then only group A would benefit from its externalities.

Case 4: Group D generates constant marginal negative externality β for all neighbours beyond threshold X and A generates constant marginal positive externality φ for all neighbours below threshold Y (where Y defined as maximum D where φ persists)

Here, from the perspective of the 'epidemic/social norm' mechanism, I combine Cases 2 and 3 to consider implications of the assumption that different household types produce countervailing externalities that ensue at different threshold points. Figure 2 again applies, with both I_D and I_A functions operative simultaneously.[7]

The outcome of the efficiency analysis rests on the relative magnitudes of the two externalities being generated. In this situation, inter-neighbourhood variations in the household mixture in the range between the two thresholds produce a constant I_T because the net marginal externality combining both functions is a constant (following the logic of Case 1). Put differently, switching group A and D households between neighbourhoods that have exceeded both thresholds (and continue to do so after the hypothetical reallocation) will lead to no net change in efficiency because the gains in the destination neighbourhood will exactly offset the losses in origin neighbourhood. However, whether such a mixed situation will be superior to more segregated options

cannot be ascertained without more information about the relative magnitudes of the two externality parameters φ and β.

Consider the following thought experiment. What would happen to efficiency were we to reallocate households within some of these mixed (i.e. with %D values between Y and X) neighbourhoods so that we instead produced more segregated neighbourhoods (i.e. with %D below both thresholds in some and above both thresholds in the others)? If the positive externality produced by group A was much greater than the negative externality produced by group D, the former set of neighbourhoods would enjoy massive increases in positive externalities associated with now-larger percentages of group A residents, with no offsetting negative externalities from group D (because their percentage would be below X). By contrast, the latter set of neighbourhoods would evince some increase in negative externalities associated with their now-larger percentages of D households and no offsetting positive externalities from group A (because their percentage would be below 100-Y). If indeed $|\varphi| > |\beta|$, then the gains in the former set of neighbourhoods will offset the losses in the other, and it will be more efficient for society as a whole to convert mixed neighbourhoods into those that have more segregation.

The conclusion is the opposite if we reverse the assumptions about the relative magnitudes of the externalities and replay our thought experiment. If $|\varphi| < |\beta|$, then the gains in the set of neighbourhoods with a now-larger share of group A will not offset the losses in the other, and it will be more efficient to avoid switching mixed neighbourhoods for those that have more segregation. It is mathematically possible, of course, that the two parameters are precisely equal in absolute value, such that all allocations are equally efficient. For a mathematical demonstration of these conclusions, see the Appendix, Case 4.

The equity analysis here proceeds analogously to Cases 2 and 3, yielding grounds for mixing. Even if potentially strong positive externalities generated by group A were extant, their implied segregation on efficiency grounds would conflict with the fairness goal of having some D households share in the positive externalities instead of solely experiencing negative ones in their neighbourhoods.

Case 5: Group D generates growing marginal negative externality for only D neighbours

The externality modelled here can be considered a 'selective socialization' process, wherein the assumed bad influence of one D household is felt only by other D households in the neighbourhood, perhaps because they are more vulnerable. Here the negative externality produced by the marginal D household increases nonlinearly with the number of D households in a given neighbourhood because there are more D neighbours to be affected by the externality. Households in group A are assumed irrelevant as either transmitters or receivers of this externality, perhaps because they have few social networks involving group D or because they have a great social distance from

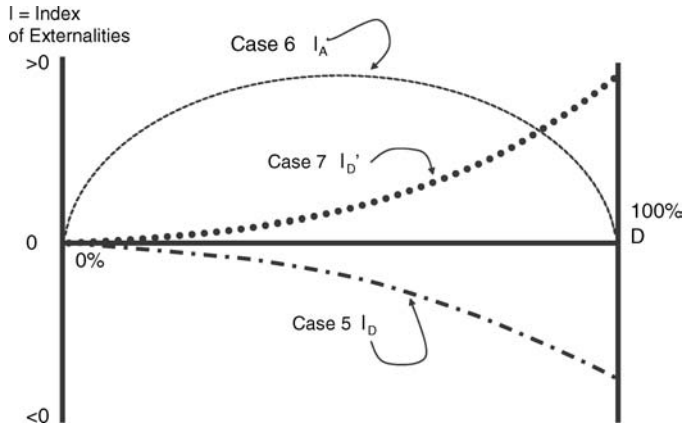


Figure 3. Case 5: D produces externality < 0 only for neighbouring D (I_D). Case 6: A produces externality > 0 only for neighbouring D (I_A). Case 7: D produces externality > 0 only for neighbouring D (I_D').

them. The nonlinear externality function for group D in one neighbourhood, I_D, is shown in Figure 3.

In this case, allocating all group D households to neighbourhoods in the smallest percentages possible, equally across all neighbourhoods, would minimize the total negative externalities that they produce for themselves. That is, because the negative externality grows more than proportionately with the addition of one more group D household, these households should be dispersed in the lowest feasible, equal concentrations for the most efficient solution.

Equity concerns produce the identical implication. To minimize their exposure to negative externalities each group D household should, ideally, reside in neighbourhoods where they are the only representative of their group.

Case 6: Group A generates growing marginal positive externality for only D neighbours

Here with the 'selective socialization' mechanism the marginal externality produced by A (φ) benefits each D present in the neighbourhood, so it can be expressed φ%D. This could represent a situation wherein each group A household provides a valuable role model for all group D households present, which is irrelevant for other group A households because they are already assumed to evince this behaviour. The corresponding I_A externality function for a particular neighbourhood is shown in Figure 3.

Efficiency concerns imply that the maximum positive externality in any neighbourhood will occur at an even mix of group A and D households. Thus the

global efficiency maximum will occur if as many neighbourhoods as feasible are mixed at a 50–50 split, until either all group A or D households have been housed.

Equity considerations reach the same conclusion. To have group D gain the most positive external benefits implies that they be evenly mixed with group A households, to the extent feasible.

Case 7: Group D generates growing marginal positive externality for only D neighbours

This variant of the 'social network' mechanism describes what might be called 'group affinity'. The notion is that as more group D households cluster in space they can build stronger social ties within the group and build valuable cultural capital. Here the total positive externality increases nonlinearly with the number of D in a given neighbourhood because there are more D neighbours to be affected by the externality. Group A households are assumed irrelevant as either transmitter or receiver of externality. The group D externality function in this case is shown as I_D' in Figure 3.

Efficiency and equity considerations lead to the opposite conclusions here compared to Case 5. In this case, allocating all D to homogeneous D-occupied neighbourhoods yields a higher value for I_T that if D were allocated in any smaller percentages across neighbourhoods. Because the marginal benefit of an added group D neighbour rises as more of group D are already present, such a household always should be added to the neighbourhood with the greatest %D, up to a maximum of 100 per cent. Analogously, this is the way to maximize the benefits accruing to group D households, without in any way detracting from the benefits accruing to group A households in their neighbourhoods.

Case 8: Group A and Group D generate growing marginal positive externalities but for only neighbours not like themselves

This variant of the 'social network' mechanism describes what might be called 'social cohesion'. In this view, there may be nothing intrinsically good or bad about the behaviours and attitudes of either group, but there is a larger societal value in the social interaction between them in a neighbourhood context. That is, mutual and equal positive externalities are generated for all participants when they reside together. The identical I_D and I_A externality functions that correspond to this case are presented in Figure 4.

From efficiency and equity perspectives, this is analogous to Case 6, wherein both A and D produce positive externalities whose marginal benefits are proportional to the other group in the area. As before, both efficiency and equity considerations yield the implication that as many neighbourhoods as possible be mixed at equal percentages of group A and D households.

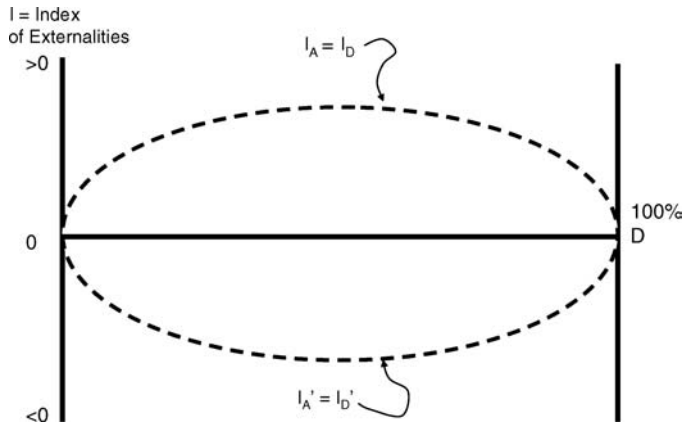


Figure 4. Case 8: A, D produces externality > 0 only for unlike neighbours. Case 9: A, D produces externality < 0 only for unlike neighbours.

Case 9: Group A and Group D generate growing marginal negative externalities but for only neighbours not like themselves

Here I model the 'competition' and 'relative deprivation' mechanisms. Each member of both groups is assumed to receive disamenities from the presence of members of the other group in the neighbourhood. Their common I_D' and I_A' externality functions are presented in Figure 4.

This is analysed as the converse of Case 8, wherein both A and D produce negative instead of positive externalities whose marginal benefits are proportional to the other group in the area. From efficiency and equity perspectives alike, one draws the conclusion that as few neighbourhoods as possible be mixed; rather group A and D households should be completely segregated.

Case 10: Group D generates constant negative externality β for all neighbours if D exceeds threshold X

This can be seen as the 'neighbourhood stigmatizing' mechanism. If the external marketplace holds stereotypical views about group D, it may develop negative responses towards anyone from a neighbourhood where group D constitutes more than X per cent of the households. Although in some sense it is not the fault of group D households that they are stereotyped in this fashion, it is appropriate to model this as if they indeed were the source of this externality, even though it is applied from outside the neighbourhood ultimately. Two alternative forms of this mechanism can be envisioned: one that has the externality negatively related to %D in a continuous fashion (see $I_A = I_D$ in Figure 5); the other with a constant externality that is imposed discontinuously once %D exceeds X (see $I_A' = I_D'$ in Figure 5).

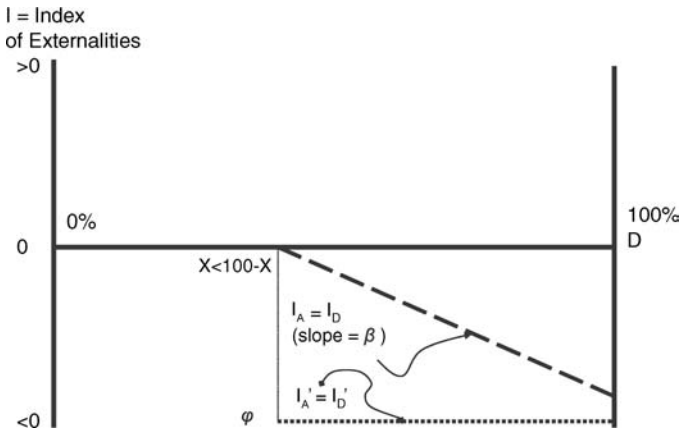


Figure 5. Case 10: neighbourhood stigmatized past threshold @X; either marginal externalty $= \beta < 0$ for all, or constant, lump-sum externality $= \varphi < 0$ for all.

From both efficiency and equity standpoints, it is clearly preferable to avoid concentrations of group D households that exceed the stigmatizing threshold X. If group D represents a small share of all households and/or X is large, this well may be mathematically feasible. In other cases, it is preferable to allocate group D in such a way that as many neighbourhoods as possible do not exceed X, with the remaining D households residing in homogeneous D-occupied neighbourhoods.

Deconstructing neighbourhood effects arising from external sources

As explained above, there is a variety of common impacts on individuals arising because they reside in the same neighbourhood and are thus subjected to the same external forces upon that neighbourhood. These mechanisms included job-housing spatial mismatch, weak local institutional resources and inferior public services.

Regardless of which external mechanism is posited, it is extremely difficult to draw any efficiency implications involving neighbourhood household mix. Such would necessarily need to make recourse to general equilibrium models of the metropolitan economy that are beyond the scope of current science. Equity issues are more easily raised, for in each case one can argue that the burden of occupying neighbourhoods with one or more of the disadvantages noted above should be borne by both household groups. This need not imply mix within individual exogenously disadvantaged neighbourhoods, of course, only that across them as a set there are different household groups represented.

A summary of the analytical results

The conclusions from the foregoing analysis can be usefully summarized in a more compact form as shown in Table 1. The main point that should become immediately

Table 1. Summary of neighbourhood mixing implications from alternative neighbourhood effect mechanisms emanating from groups A, D (shown in brackets)

Neighbourhood effect type	Equity argument	Efficiency argument
1. $D < 0$, $A > 0$; constant [Socialization]	Mix	Alternative allocations make no difference
2. $D < 0$; threshold [Epidemic/social norm]	Mix with D not exceeding threshold	Mix with D not exceeding threshold anywhere; 100%D for remainder if necessary
3. $A > 0$; threshold [Epidemic/social norm]	Mix with A exceeding threshold	Mix with A exceeding threshold everywhere; if impossible, do 100% A until A exhausted
4. $D < 0$; threshold $A > 0$; threshold [Epidemic/social norm]	Mix in range between two thresholds	Complete segregation if A's externality > D's; mix in range between two thresholds if D's externality > A's
5. $D < 0$ other D; proportional [Selective socialization]	D dispersed to lowest feasible equal % everywhere	D dispersed to lowest feasible equal % everywhere
6. $A > 0$ for D; proportional [Selective socialization]	Equal % A and D wherever D present	Equal % A and D wherever D present, if possible; mix in remainder irrelevant
7. $D > 0$ other D; proportional [Social networks]	Complete segregation of D	Complete segregation of D wherever possible; highest % D elsewhere
8. D and $A > 0$ for others; proportional [Social networks]	Equal % A and D wherever D present	Equal % A, D wherever D present, if possible; mix in remainder irrelevant
9. D and $A < 0$ for others; proportional [Competition; relative deprivation]	Complete segregation of A and D	Complete segregation of A and D
10. $D < 0$; threshold [Stigmatization]	Mix with D not exceeding threshold	Mix with D not exceeding threshold anywhere; 100% D for remainders if necessary
11. External forces [Spatial mismatch, local institutional resources and public services]	Mix weak neighbourhoods or spread groups among same	Indeterminate

obvious is that *virtually every different purported mechanism of neighbourhood effect implies on efficiency and/or equity grounds a distinctive optimal pattern of neighbourhood mixing of households*. These run the gamut, including complete segregation of groups, minimization of group D representation everywhere, achieving minimum amounts of group A everywhere, mixing of A and D groups within a potentially broad band in the realm of dual thresholds, and precisely even allocations.

Still other mechanisms imply that neighbourhood mix is irrelevant on efficiency grounds.

Conclusions and Implications for Neighbourhood Mix Policies in western Europe

A large variety of mechanisms has been advanced in the scholarly literature for how differences in the neighbourhood environment may affect the social outcomes of individual residents. Most importantly for the current policy context in western Europe, the majority of these mechanisms are internal to the neighbourhood, i.e. they reputedly emanate from the household mix. But precisely *how and why* neighbourhoods matter must be unpacked carefully before one can leap to any policy implications regarding neighbourhood mixing.[8] Toward this end, I have analysed comprehensively, in theoretical terms, the alternative mechanisms for how neighbourhood effects might occur, showing that different mechanisms lead to radically different conclusions regarding desired neighbourhood household mix on either equity and/or efficiency grounds.

This result means that even the crudest guidance for policy aimed at achieving an optimal mix of households among neighbourhoods depends on the careful, explicit delineation of precisely which mechanisms of neighbourhood effects are operative, and perhaps the relative magnitudes of the externalities involved if multiple effects are operative. By implication, information on what sorts of social externality processes actually *are occurring* in their nation's neighbourhoods must be of paramount importance to policymakers. Clearly, a valuable next step would be to conduct a meta-analysis of the western European research evidence in this regard, as has been suggested by Tunstall and Fenton (2006). I hope that this paper serves to motivate and frame this vital exercise. Until then, the common policy thrust toward neighbourhood social mixing must be seen as based more on faith than fact.

Acknowledgements

In preparing this document I have benefited greatly from the assistance provided by Sako Musterd and Wim Ostendorf, University of Amsterdam, and Roger Andersson, Uppsala University. I also express gratitude for the excellent clerical support provided by Noelia Caraballo and Phyllis Seals. Two anonymous referees provided helpful suggestions, but the opinions (and potential errors) contained in this document are my own.

Notes

1. Some analysts (notably Friedrichs, 1998) have denoted 'exposure to crime and violence' as a distinct neighbourhood factor. Indeed, there is evidence that this aspect of neighbourhoods is associated with a variety of health outcomes for residents (see Ellen & Turner, 2003, for a review). Nevertheless, as I will show later, crime and violence are themselves related to the more fundamental neighbourhood

social processes that I list here. I therefore do not list it as a separate class of neighbourhood effect mechanism.

2. Over time, income certainly could change. Indeed, several of the potential intra-neighbourhood externalities would be expected to affect the incomes of A or D residents; it is the nature of the externality itself.
3. Although we know this factually is untrue, policymakers make this same simplifying assumption when justifying mixing strategies.
4. Note that this threshold is expressed here in proportionate terms, which is theoretically equivalent to an absolute number given the simplifying assumption of a fixed neighbourhood population. In empirical work, however, identifying whether the threshold is based on an absolute or proportionate number of a group is critical.
5. More modern sociological treatises closely related to collective socialization also suggest thresholds, such as Wilson's (1987) contention that as a critical mass of middle-class families leave the inner-city, low-income blacks left behind become isolated from the positive role models that the erstwhile dominant class offered. Economists also have developed several mathematical treatises involving collective socialization effects in which thresholds often emerge as solutions to complex decision problems under certain assumptions (Akerlof, 1980; Galster, 1987, Ch. 3; Brock & Durlauf, 2001).
6. There is no necessary reason why past the threshold the relationship is linear; this is done for simplicity here.
7. Figure 2 portrays $Y > X$, but this is not necessary and the textual discussion does not depend on this.
8. For a vigorous debate on the policy implications of neighbourhood effects research, see Atkinson & Kintrea (2002), Friedrichs (2002), Kearns (2002), Musterd (2002, 2003), and the set of responses to McCulloch (2001) in the same issue of *Environment and Planning A*., pp. 1335–1369. Of course, this discussion presumes that government policies can have some non-trivial influence over neighbourhood social mix, a presumption that is subject to question.

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Appendix: Mathematical Analyses of Most Efficient Social Mix given Different Neighbourhood Externality Effects

This appendix provides some rigour to the graphical and heuristic arguments made in the text. It is based on the same simplifying assumptions and symbolism explained in the text and proceeds in a parallel organizational fashion. The goal is to ascertain which mix in both stylized neighbourhoods maximizes I_T (or minimizes it when it is negative at all mixes).

Case 1: D Generates Constant Marginal Negative Externality β for All Neighbours, A Generates Constant Marginal Positive Externality φ for All Neighbours

$$I_D = \beta(\%D) + \beta(100\text{-}\%D) = 100\beta \qquad \beta < 0$$

$$I_A = 100\varphi - \varphi(\%D) + 100\varphi - \varphi(100 - \%D) = 100\varphi \qquad \varphi > 0$$

$$I_T = I_D + I_A = 100(\beta + \varphi)$$

and because I_T is not a function of %D it will not vary by neighbourhood mix; I_T is constant across all mixes. Whether I_T is positive or negative depends on relative magnitudes of β and φ.

Case 2: Group D Generates Constant Marginal Negative Externality β for All Neighbours Beyond Threshold X

$$
\begin{aligned}
I_D &= \beta(100\text{-}\%D\text{-}X) \quad \text{if } \%D \leq X; \\
&= \beta(\%D - X) + \beta(100 - \%D - X) = (100 - 2X)\beta \quad \text{if } X < \%D \leq 100 - X \\
&= \beta(\%D - X) \text{ if } \%D > 100 - X \qquad\qquad \beta < 0 \\
I_A &= 0 \\
I_T &= I_D + I_A = I_D
\end{aligned}
$$

so we minimize by considering each segment separately. When $\%D \leq X$ I_T is minimized ($= 100\beta$) by making $\%D = X$; when $X < \%D \leq 100\text{-}X$ I_T is a constant ($= (100 - 2X)\beta$); when $\%D > 100\text{-}X$ I_T is minimized by making %D as small as possible ($\approx 100\text{-}X$), rendering $I_T = (100 - 2X)\beta$. Thus, I_T overall is minimized by allocating X% D households in one neighbourhood and 100-X% in the other.

Case 3: Group A Generates Constant Marginal Positive Externality φ for All Neighbours Beyond Threshold Y (Where Y Defined as Maximum D Where φ Persists)

$$
\begin{aligned}
I_A &= Y\varphi - \varphi(\%D) \text{ if } \%D \leq 100 - Y; \\
&= Y\varphi - \varphi(\%D) + Y\varphi - \varphi(100 - \%D) = 2Y\varphi - 100\varphi \quad \text{if } 100 - Y < \%D \leq Y \\
&= Y\varphi - \varphi(100 - \%D) \text{ if } \%D > Y \qquad\qquad \varphi > 0 \\
I_D &= 0 \\
I_T &= I_D + I_A = I_A
\end{aligned}
$$

so we minimize by considering each A segment separately. When $\%D \leq 100\text{-}Y$ I_T is maximized ($= Y\varphi$) by making $\%D = 0$ (i.e. $dI_T/d\%D < 0$); when $100\text{-}Y < \%D \leq Y$, I_T is a constant ($= 2Y\varphi - 100\varphi$); when $\%D > Y$, I_T is maximized ($= Y\varphi$) by making %D as large as possible (i.e. $dI_T/d\%D > 0$); rendering $I_T = Y\varphi$. Note that since $Y < 100$, $2Y\varphi - 100\varphi < Y\varphi$. Thus, I_T overall is maximized by allocating 100% D households in one neighbourhood and none in the other.

Case 4: Group D Generates Constant Marginal Negative Externality β for All Neighbours Beyond Threshold X and A Generates Constant Marginal Positive Externality φ for All Neighbours Below Threshold Y (Where Y Expressed in Terms of %D and, for Simplicity, Y = 100-X)

If $x < Y$:

$$
\begin{aligned}
I_D &= \beta(100 - \%D - X) \text{ if } \%D \leq X; \\
&= \beta(\%D - X) + \beta(100 - \%D - X) = \beta(100 - 2X) \text{ if } X < \%D \leq 100 - X = Y \\
&= \beta(\%D - X) \text{ if } \%D > Y \qquad\qquad \beta < 0
\end{aligned}
$$

$$
\begin{aligned}
I_A &= Y\varphi - \varphi(\%D) \text{ if } \%D \leq 100 - Y = X\ ; \\
&= Y\varphi - \varphi(\%D) + Y\varphi - \varphi(100 - \%D) = 2Y\varphi - 100\varphi \text{ if } X < \%D \leq Y \\
&= Y\varphi - \varphi(100 \text{ - } \%D) \text{ if } \%D > Y \qquad\qquad \varphi > 0 \\
I_T &= I_D + I_A \\
&= \beta(100 - \%D \text{ - } X) + Y\varphi - \varphi(\%D) \text{ if } \%D \leq X; \\
&= \beta(100 - 2X) + 2Y\varphi - 100\varphi \text{ if } X < \%D \leq Y \\
&= \beta(\%D - X) + Y\varphi - \varphi(100 - \%D) \text{ if } \%D > Y
\end{aligned}
$$

If $X > Y$:

$$
\begin{aligned}
I_D &= \beta(100 - \%D - X) \text{ if } \%D \leq 100 - X = Y; \\
&= 0 \text{ if } Y < \%D \leq X \\
&= \beta(\%D \text{ - } X) \text{ if } \%D > X\ \beta < 0 \\
I_A &= Y\varphi - \varphi(\%D) \text{ if } \%D \leq Y; \\
&= 0 \text{ if } Y < \%D \leq X \\
&= Y\varphi - \varphi(100 - \%D) \text{ if } \%D > X\ \varphi > 0 \\
I_T &= I_D + I_A \\
&= \beta(100 - \%D - X) + Y\varphi - \varphi(\%D) \text{ if } \%D \leq Y; \\
&= 0 \text{ if } Y < \%D \leq X \\
&= \beta(\%D - X) + Y\varphi - \varphi(100 - \%D) \text{ if } \%D > X
\end{aligned}
$$

Now $dI_T/d\%D = 0$ always when $X < \%D \leq Y$, so society is indifferent to mixes within this range X–Y. However, if $|\varphi| > |\beta|$, $dI_T/d\%D$ will be < 0 when $\%D \leq$ the lower threshold, and $dI_T/d\%D$ will be $>$ 0 when $\%D >$ the higher threshold; the signs will be reversed if the relative magnitudes of φ and β are reversed. Note also that $I_T(\%D = 0) = I_T(\%D = 100) = Y\varphi + \beta(100 - X)$; the former is > 0, the latter $<$ 0. Thus, the only choices for optimum are between 100% D in one neighbourhood-zero in the other, or two with mixes in the range of X–Y. Which will be preferred cannot be ascertained without more information about the relative magnitudes of φ and β. If the positive externality from A is much more powerful than the negative externality from D, i.e. if $|\varphi| > |\beta|$, then the mixed option will be inferior. On the contrary, if the negative externality from D is much more powerful than the positive externality from A, i.e. if $|\varphi| < |\beta|$, then the mixed option will be the maximum I_T available. It is mathematically possible, of course, that the two parameters are precisely equal in absolute value, such that all allocations are equally efficient.

Case 5: Group D Generates Growing Marginal Negative Externality for Only D Neighbours

Here the marginal negative externality increases with the number of D in a given neighbourhood ($= \beta\%D$) because there are more D neighbours to be affected by the externality. The total externality function for D in that neighbourhood thus can be expressed $\beta\%D^2/2$. Group A are assumed irrelevant as either transmitter or receiver of externality. For both neighbourhoods:

$$
\begin{aligned}
I_D &= \beta\%D^2/2 + \beta(100 - \%D)^2/2 \qquad\qquad \beta < 0 \\
&= 5,000\beta - 100\beta\%D + \beta\%D^2
\end{aligned}
$$

$$I_A = 0$$

$$I_T = I_D + I_A = I_D = 5,000\beta - 100\beta\%D + \beta\%D^2$$

In this case, allocating all D to one neighbourhood or the other yields $I_T = 5{,}000\beta$, which is a lower (more negative) value that if D were equally allocated in the smallest possible percentage across all neighbourhoods (here, 50 per cent), whereupon $I_T = 2{,}500\beta$. Expressed differently, $dI_T/d\%D = -100\beta + 2\beta\%D$, which is minimized at $\%D = 50$.

Case 6: Group D Generates Growing Marginal Positive Externality for Only D Neighbours

Here the marginal positive externality increases with the number of D in a given neighbourhood ($= \beta\%D$) because there are more D neighbours to be affected by the externality. The total externality function for D in that neighbourhood thus can be expressed $\beta\%D^2/2$. Group A are assumed irrelevant as either transmitter or receiver of externality. For both neighbourhoods, therefore:

$$I_D = \beta\%D^2/2 + \beta(100 - \%D)^2/2 \qquad \beta > 0$$

$$= 5,000\beta - 100\beta\%D + \beta\%D^2$$

$$I_A = 0$$

$$I_T = I_D + I_A = I_D = 5,000\beta - 100\beta\%D + \beta\%D^2$$

In this case, allocating all D to one neighbourhood or the other yields $I_T = 5{,}000\beta$, which is a higher (positive) value that if D were equally allocated in the smallest possible percentage across all neighbourhoods (here, 50 per cent), whereupon $I_T = 2{,}500\beta$.

Case 7: Group A Generates Growing Marginal Positive Externality for Only D Neighbours

Here the marginal externality produced by A (φ) benefits each D present in the neighbourhood, so it can be expressed $\phi\%D$. The total externalities produced in one neighbourhood by A is thus $\varphi\%DA$, or $\varphi\%D(100\text{-}\%D) = 100\varphi\%D - \varphi\%D^2$. Analogously, in the other neighbourhood the total externalities produced by A will be: $100\varphi(100\text{-}\%D) - \varphi(100\text{-}\%D)^2$. After simplification we can write:

$$I_D = 0$$

$$I_A = 200\varphi\%D - 2\varphi\%D^2 \qquad \varphi > 0$$

$$I_T = I_D + I_A = I_A = 200\varphi\%D - 2\varphi\%D^2$$

The maximum of I_T occurs here at a 50–50 split of D between neighbourhoods, as can be seen by setting $dI_T/d\%D$ ($= 200\varphi - 4\varphi\%D$) to zero.

Case 8: Group A and Group D Generate Growing Marginal Positive Externalities but for Only Neighbours Not Like Themselves

This is analogous formally to Case 6, wherein both A and D produce positive externalities whose marginal benefits are proportional to the other group in the area, i.e.

$$I_D = 200\beta\%D - 2\beta\%D^2 \qquad \beta > 0$$

$$I_A = 200\varphi\%D - 2\varphi\%D^2 \qquad \varphi > 0$$

$$I_T = I_D + I_A = 200\beta\%D - 2\beta\%D^2 + 200\varphi\%D - 2\varphi\%D^2$$
$$= 200(\beta + \varphi)\%D - 2(\beta + \varphi)\%D^2$$

As in Case 7, the maximum of I_T occurs here at a 50–50 split of D between neighbourhoods, as can be seen by setting $dI_T/d\%D$ $(= 200(\beta + \varphi) - 4(\beta + \varphi)\%D)$ to zero.

Case 9: Group A and Group D Generate Growing Marginal Negative Externalities but for Only Neighbours Not Like Themselves

This is analogous formally to Case 8, wherein both A and D produce negative instead of positive externalities whose marginal benefits are proportional to the other group in the area, i.e.

$$I_D = 200\beta D - 2\beta\%D^2 \qquad \beta < 0$$
$$I_A = 200\varphi D - 2\varphi\%D^2 \qquad \varphi < 0$$
$$I_T = I_D + I_A = 200\beta\%D - 2\beta\%D^2 + 200\varphi\%D - 2\varphi\%D^2$$
$$= 200(\beta + \varphi)\%D - 2(\beta + \varphi)\%D^2$$

The minimum of I_T occurs here at a 50–50 split of D between neighbourhoods, as can be seen by setting $dI_T/d\%D$ $(= 200(\beta + \varphi) - 4(\beta + \varphi)\%D)$ to zero. Thus, to avoid this low I_T situation, group A and D households should be completely segregated.

Case 10: Group D Generates Constant Negative Externality β for All Neighbours if D Exceeds Threshold X

If X < 100-X, then no matter how D is allocated in this simplified situation at least one neighbourhood will exceed X and the negative externality will result, i.e. the externality functions can be specified:

$$I_D = I_A$$
$$= \beta \text{ if } \%D \leq X;$$
$$= 2\beta \text{ if } X < \%D \leq 100 - X$$
$$= \beta \text{ if } \%D > 100 - X \qquad \beta < 0$$
$$I_T = I_D + I_A =$$
$$= 2\beta \text{ if } \%D \leq X;$$
$$= 4\beta \text{ if } X < \%D \leq 100 - X$$
$$= 2\beta \text{ if } \%D > 100 - X$$

In this case the reductions in I_T can be minimized by allocating D such that one neighbourhood is below the threshold, i.e. $\%D \leq X$.

If X > 100-X, the externality function becomes:

$$I_D = I_A$$
$$= \beta \text{ if } \%D \leq 100\text{-}X;$$
$$= 0 \text{ if } 100\text{-}X < \%D \leq X$$
$$= \beta \text{ if } \%D > X \qquad \beta < 0$$

$$
\begin{aligned}
I_T &= I_D + I_A = \\
&= 2\beta \text{ if } \%D \leq \text{100-X;} \\
&= 0 \text{ if 100-X} < \%D \leq X \\
&= 2\beta \text{ if } \%D > X
\end{aligned}
$$

In this case the reductions in I_T can be avoided altogether by allocating D such that both neighbourhoods are below the threshold, i.e. if 100-X $< \%D \leq$ X.